MULTIVARIABLE CALCULUS • UNIT 1.01

C Functions of Several Variables, Limits & Partial Derivatives

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

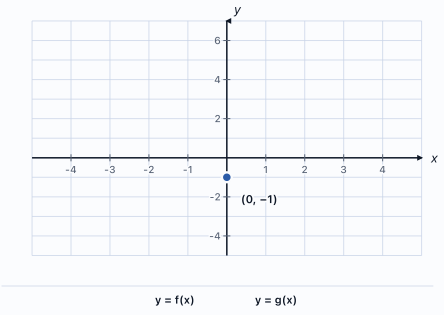
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 B	Q3 C	Q4 A	Q5 A
Q6 B	Q7 D	Q8 A	Q9 C	Q10 B
Q11 A	Q12 A	Q13 B	Q14 B	Q15 B
Q16 A	Q17 C	Q18 D	Q19 A	Q20 A
Q21 D	Q22 C	Q23 D	Q24 B	Q25 B

Functions of Several Variables, Limits & Partial Derivatives Practice Exercises

1 MCQ

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.



A 10

B 13

C 26 ✓

D 52

WORKED SOLUTION & VERIFICATION

We are given $z = f(x, y) = x^2 + y^2$, with $x = 2t$ and $y = 3t$. We need to find $\frac{dz}{dt}$ using the Multivariable Chain Rule:
$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$.
First, find the partial derivatives of z with respect to x and y:
$\frac{\partial z}{\partial x} = 2x$
$\frac{\partial z}{\partial y} = 2y$
Next, find the derivatives of x and y with respect to t:
$\frac{dx}{dt} = 2$
$\frac{dy}{dt} = 3$
Now, substitute these into the chain rule formula:
$\frac{dz}{dt} = (2x)(2) + (2y)(3) = 4x + 6y$.
We are asked to find $\frac{dz}{dt}$ when $t = 1$. First, find the values of x and y when $t = 1$:
$x = 2(1) = 2$
$y = 3(1) = 3$
Finally, substitute these values of x and y into the expression for $\frac{dz}{dt}$:
$\left.\frac{dz}{dt}\right|_{t=1} = 4(2) + 6(3) = 8 + 18 = 26.$

2 MCQ

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

A 12	B 18 ✓
C 6	D 24

WORKED SOLUTION & VERIFICATION

To evaluate the double integral $\iint_R 2xy\, dA$ where $R = [0, 2] \times [0, 3]$, we can use Fubini's Theorem. We will set up the iterated integral by integrating with respect to y first, then x:

$$\int_0^2 \int_0^3 2xy\, dy\, dx$$

First, integrate with respect to y, treating x as a constant:

$$\int_0^3 2xy\, dy = [xy^2]_{y=0}^{y=3} = x(3^2) - x(0^2) = 9x$$

Now, integrate the result with respect to x:

$$\int_0^2 9x\, dx = [\frac{9x^2}{2}]_{x=0}^{x=2} = \frac{9(2^2)}{2} - \frac{9(0^2)}{2} = \frac{9(4)}{2} = \frac{36}{2} = 18$$

Therefore, the value of the double integral is 18.

3 MCQ

For $f(x, y) = 7x^2 + 4y^2$, find $f_y(-3, 1)$.

A -24	B 4
C 8 ✓	D -42

WORKED SOLUTION & VERIFICATION

Applying Partial Derivatives principles, $f_y = 8y$, so $f_y(-3, 1) = 8(1) = 8$. Therefore, the verified correct option is 8.

4 MCQ

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

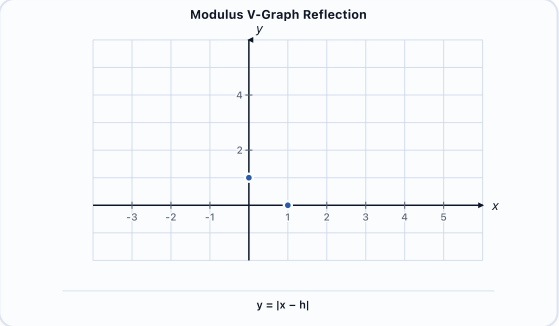
A 2 ✓	B 4
C 6	D 0

WORKED SOLUTION & VERIFICATION

First, we find the first partial derivative of $f(x, y)$ with respect to x: $f_x(x, y) = \frac{\partial}{\partial x}(x^2 + y^2) = 2x$.
Next, we find the second partial derivative of $f(x, y)$ with respect to x (which is f_{xx}): $f_{xx}(x, y) = \frac{\partial}{\partial x}(f_x(x, y)) = \frac{\partial}{\partial x}(2x) = 2$.
Finally, we evaluate $f_{xx}(x, y)$ at the point $(2, 3)$. Since $f_{xx}(x, y)$ is a constant, its value at any point is 2. Therefore, $f_{xx}(2, 3) = 2$.

5 MCQ

Let the linear scalar potential be $h(u, y) = 15u + 8y - 16$. Compute the squared gradient norm $\|\nabla h\|^2$.



- **A** 289 ✓
- **B** 293
- **C** 281
- **D** 304

WORKED SOLUTION & VERIFICATION

Since $\nabla h = \langle 15, 8 \rangle$, $\|\nabla h\|^2 = 15^2 + 8^2 = 289$.

6 MCQ

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

- **A** $\sqrt{14}$
- **B** $2\sqrt{14}$ ✓
- **C** $3\sqrt{14}$
- **D** $4\sqrt{14}$

WORKED SOLUTION & VERIFICATION

The surface area of a surface $z = g(x, y)$ above a region R is given by the integral $\iint_R \sqrt{1 + (g_x)^2 + (g_y)^2}\, dA$. Here, $g(x, y) = 2x + 3y + 1$. The partial derivatives are $g_x = 2$ and $g_y = 3$. Therefore, $\sqrt{1 + (g_x)^2 + (g_y)^2} = \sqrt{1 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$. The integral becomes $\iint_R \sqrt{14}\, dA$, where $R = [0, 1] \times [0, 2]$. This evaluates to $\sqrt{14} \times \text{Area}(R)$. The area of the rectangle R is $(1 - 0) \times (2 - 0) = 2$. So, the surface area is $2\sqrt{14}$.

7 MCQ

Consider the vector field $\mathbf{u}(s, y, z) = \langle 22s,\ 7y,\ 7z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{u}$.

- **A** 58
- **B** 40
- **C** 29
- **D** 36 ✓

WORKED SOLUTION & VERIFICATION

The divergence is $22 + 7 + 7 = 36$.

8 MCQ

Consider the vector field $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. Evaluate the vector field at the point $(2, 4)$.

- **A** $\langle 6, 8 \rangle$ ✓
- **B** $\langle 8, 6 \rangle$
- **C** $\langle 5, 6 \rangle$
- **D** $\langle 6, 5 \rangle$

WORKED SOLUTION & VERIFICATION

The vector field is given by $\mathbf{F}(x, y) = \langle 3x, 2y \rangle$. To evaluate the vector field at the point $(2, 4)$, we substitute $x = 2$ and $y = 4$ into the components of the vector field.
First component: $3x = 3(2) = 6$.
Second component: $2y = 2(4) = 8$.
Therefore, $\mathbf{F}(2, 4) = \langle 6, 8 \rangle$.

9 MCQ

A particle's position in three-dimensional space is given by the vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, where t represents time. The velocity of the particle is given by $\mathbf{v}(t) = \mathbf{r}'(t)$. Suppose the velocity components are related by the partial differential equations: $\frac{\partial v_x}{\partial t} = -2x(t) + 3y(t)$ and $\frac{\partial v_y}{\partial t} = x(t) - 4y(t)$. If at time $t = 0$, the particle is at the origin $(0, 0, 0)$ and its velocity is $\mathbf{v}(0) = \langle 5, -2, 0 \rangle$, determine the value of $\frac{\partial^2 v_x}{\partial t^2}$ at $t = 0$.

Given Vector Components

Component x	Component y	Component z
0	0	0

- **A** 16
- **B** 26
- **C** -16 ✓
- **D** -26

WORKED SOLUTION & VERIFICATION

We are given the partial differential equations relating the velocity components: $\frac{\partial v_x}{\partial t} = -2x(t) + 3y(t)$ and $\frac{\partial v_y}{\partial t} = x(t) - 4y(t)$. To find $\frac{\partial^2 v_x}{\partial t^2}$, we differentiate the first equation with respect to t: $\frac{\partial^2 v_x}{\partial t^2} = \frac{\partial}{\partial t}(-2x(t) + 3y(t))$. Using the chain rule, this becomes $\frac{\partial^2 v_x}{\partial t^2} = -2\frac{\partial x}{\partial t} + 3\frac{\partial y}{\partial t}$. Since $\mathbf{v}(t) = \langle v_x(t), v_y(t), v_z(t) \rangle = \langle \frac{\partial x}{\partial t}, \frac{\partial y}{\partial t}, \frac{\partial z}{\partial t} \rangle$, we have $\frac{\partial x}{\partial t} = v_x(t)$ and $\frac{\partial y}{\partial t} = v_y(t)$. Therefore, $\frac{\partial^2 v_x}{\partial t^2} = -2v_x(t) + 3v_y(t)$. We are given that at $t = 0$, $\mathbf{v}(0) = \langle 5, -2, 0 \rangle$, so $v_x(0) = 5$ and $v_y(0) = -2$. Substituting these values, we get: $\frac{\partial^2 v_x}{\partial t^2}\Big|_{t=0} = -2(5) + 3(-2) = -10 - 6 = -16$.

10 MCQ

Consider the function $f(x, y) = 5x^3 - 2y^2 + 7$. Compute the value of the partial derivative $f_y(2, -3)$.

- A -12
- B **12** ✓
- C -20
- D 20

WORKED SOLUTION & VERIFICATION

To find the partial derivative of $f(x, y)$ with respect to y, we treat x as a constant.
Given $f(x, y) = 5x^3 - 2y^2 + 7$.
The partial derivative with respect to y is denoted by f_y.
$f_y = \frac{\partial}{\partial y}(5x^3 - 2y^2 + 7)$
Applying the power rule and noting that $5x^3$ and 7 are constants with respect to y:
$f_y = 0 - 2(2y) + 0$
$f_y = -4y$
Now, we need to evaluate f_y at the point $(2, -3)$:
$f_y(2, -3) = -4(-3)$
$f_y(2, -3) = 12$

11 MCQ

Let $X = [0, 8] \times [0, 15]$ be a rectangle in the (u, y)-plane. Compute the double integral $\iint_X 44uy \, dA$.

- A **158400** ✓
- B 158404
- C 158411
- D 158376

WORKED SOLUTION & VERIFICATION

Integrating gives $11(8^2)(15^2) = 158400$.

12 MCQ

For $f(x, y) = 5x^2 + y^2$, find $f_y(2, -1)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	5	1	6

- A **-2** ✓
- B -1
- C 20
- D 4

WORKED SOLUTION & VERIFICATION

$f_y = 2y$, so $f_y(2, -1) = 2(-1) = -2$.

13 MCQ

For $f(x, y) = 3x^2 + 6y^2$, find $f_x(-6, 1)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	3	6	9

A 6

B **-36** ✓

C -18

D 12

WORKED SOLUTION & VERIFICATION

$f_x = 6x$, so $f_x(-6, 1) = 6(-6) = -36$.

14 MCQ

For $f(x, y) = 8x^2 + 4y^2$, find $f_y(-5, 5)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	8	4	12

A -40

B **40** ✓

C 20

D -80

WORKED SOLUTION & VERIFICATION

$f_y = 8y$, so $f_y(-5, 5) = 8(5) = 40$.

15 MCQ

For $f(x, y) = 5x^2 + 7y^2$, find $f_y(-5, 5)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	5	7	12

A -70

B **70** ✓

C 35

D -50

WORKED SOLUTION & VERIFICATION

$f_y = 14y$, so $f_y(-5, 5) = 14(5) = 70$.

16 MCQ

For $f(x, y) = 8x^2 + 9y^2$, find $f_x(-4, -3)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	8	9	17

A **-64** ✓

B -48

C -32

D -54

WORKED SOLUTION & VERIFICATION

$f_x = 16x$, so $f_x(-4, -3) = 16(-4) = -64$.

17 MCQ

For $f(x, y) = x^2 + 9y^2$, find $f_x(-3, -1)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	1	9	10

A -2

B -3

C **-6** ✓

D -18

WORKED SOLUTION & VERIFICATION

$f_x = 2x$, so $f_x(-3, -1) = 2(-3) = -6$.

18 MCQ

For $f(x, y) = x^2 + 3y^2$, find $f_y(2, 4)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	1	3	4

A 12

B 4

C -24

D **24** ✓

WORKED SOLUTION & VERIFICATION

Applying Partial Derivatives principles, $f_y = 6y$, so $f_y(2, 4) = 6(4) = 24$. Therefore, the verified correct option is 24.

19 MCQ

For $f(x, y) = 8x^2 + 7y^2$, find $f_x(1, -4)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	8	7	15

- **A** 16 ✓
- **B** -64
- **C** 8
- **D** -56

WORKED SOLUTION & VERIFICATION

$f_x = 16x$, so $f_x(1, -4) = 16(1) = 16$.

20 MCQ

For $f(x, y) = 4x^2 + y^2$, find $f_y(-3, 4)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	1	5

- **A** 8 ✓
- **B** -6
- **C** 4
- **D** -24

WORKED SOLUTION & VERIFICATION

Applying Partial Derivatives principles, $f_y = 2y$, so $f_y(-3, 4) = 2(4) = 8$. Therefore, the verified correct option is 8.

21 MCQ

For $f(x, y) = 2x^2 + 7y^2$, find $f_x(-3, 2)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	7	9

- **A** 8
- **B** -6
- **C** 28
- **D** -12 ✓

WORKED SOLUTION & VERIFICATION

$f_x = 4x$, so $f_x(-3, 2) = 4(-3) = -12$.

22 MCQ

For $f(x, y) = 4x^2 + 7y^2$, find $f_x(-5, -2)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	7	11

A -16

B -20

C **-40** ✓

D -28

WORKED SOLUTION & VERIFICATION

$f_x = 8x$, so $f_x(-5, -2) = 8(-5) = -40$.

23 MCQ

For $f(x, y) = 2x^2 + 3y^2$, find $f_x(-4, 3)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	3	5

A 12

B -8

C 18

D **-16** ✓

WORKED SOLUTION & VERIFICATION

$f_x = 4x$, so $f_x(-4, 3) = 4(-4) = -16$.

24 MCQ

For $f(x, y) = 4x^2 + 7y^2$, find $f_y(-2, -6)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	4	7	11

A -28

B **-84** ✓

C -42

D -16

WORKED SOLUTION & VERIFICATION

$f_y = 14y$, so $f_y(-2, -6) = 14(-6) = -84$.

25 MCQ

For $f(x, y) = 2x^2 + 6y^2$, find $f_y(5, 1)$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	6	8

A 60

B 12 ✓

C 6

D 20

WORKED SOLUTION & VERIFICATION

$f_y = 12y$, so $f_y(5, 1) = 12(1) = 12$.